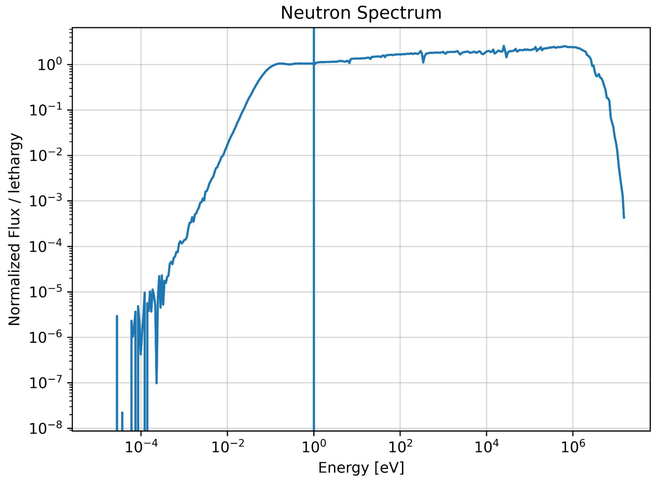

Neutron Spectrum
Normalized Flux / lethargy
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-4}
10^{-2}
10^{0}
10^{2}
10^{4}
10^{6}
Energy [eV]

Abstract:

Nuclear capable nations have been looking towards Generation IV advanced reactors to power their nuclear fleet, especially in the United States as the reactors built in the 70's and 80's are approaching the ends of their leases. Many criteria exist to be considered an 'advanced reactor' however, advanced reactors are typically characterized by the rejection of traditional reactor designs, notably water based coolant and assembly rod style fuels. A common alternative to the fuel rod style asseymblies in advanced reactors are tri structural isotropic (TRISO) particle based fuel, where TRISO particles are embedded in graphite moderated spheres, referred to as triso-pebbles. Triso fuel has higher fuel burnup than assembly rod fuels and Pebble Bed (PB) style reactors advertise a pebble circulation system, pushing these reactor designs going decades without need for refueling.

This work is motivated by the upcoming production of pebble bed reactors within the united states without the proper construction of fuel containment casks for these types of pebbles. Work has been done (Paquette at el). To study the behavior of apporpereating contemporary fuel cask designs to accommodate these spheres by stacking depleted pebbles inside the Multi Purpose Canister and filling the void space with high conductivity metals. However, this design raises issues in terms of retrieving the spent fuel for recycling when such technology exists. I propose a novel design that would make storage and retrieval of the these pebbles more robust for future use. This is achieved by using a corrugated, isometric structure of S316 steel doped with boron to hexagonally pack the depleted pebbles in a uniform and repeating pattern. For mental clarity, this design was inspired by egg-cartons.

The Xe-100 is a Pebble Bed (PB), Small Module Reactor (SMR), High Temperature Gas Reactor (HTGR) designed by X-Energy that uses tristructural isotropic (TRISO) particle fuel embedded in graphite moderated spheres. These reactors are advertised to be 'supersafe' and meltdown proof during transient operating scenarios due to their use of feedback coefficients which self regulate the neutronics within the system. This work leverages the depleted pebble concentrations modeled from Richter, with OpenMC criticality simulations to find k_eff of the novel cask design's system to determine if such a design passes the NRC's hypothetical accident conditions in regards to the neutronics. The application of this work are a novel storage system that can be retro-fitted to fit into current cask designs for pebble bed use by checking if the the neutroics pass the NRC's HAC scenario of K_eff < 0.95 with 95% certainty.

NCT

#Neutronics characteristics of a 165 MWth Xe-100 reactor, Mulder et al

	OpenMC
Version	OpenMC(TD) Ver. 2.1 [21]
Cross sections library	ENDF/B-VIII.0 [24]
Particles per batch/hist. per cycle	50,000
Batches/cycles	3,000
Inactive batches/cycles	300
Initial source position[a]	x,y,z = 0, 0, 1 cm
Initial source angular distribution	Isotropic
Boundary condition(s)	Vacuum (periodic in lattice test)

Part(s)	Material	Specifications and/or reference
Spent Fuel	Spent UO2 based fuel	initial 4.5 wt%^{235}U, 50 GWd/MTU [34]
Fresh fuel	Uranium dioxide	4.5 wt%^{235}U
Spent Inner rod gas	Helium	[26]
Fresh Inner rod gas	Fission gas	[35]
Fuel rod cladding & spacers	M5	with maximum impurity concentration [36]
Assembly empty space filling	Argon	natural abundance composition, 1 atm [6,37]
Nozzle	Steel & argon	S355JR EN10025-2:2004, 50 v.% each [38]
Profile	Steel	EN 10210 S355J2H [38]
Insert	Cast iron	EN(−1563)-GJS-400-15(GGG40) [39]
Insert lids	Steel	S355JR EN10025-2:2004 [40]
Overpack	Oxygen free copper	C102 standard [41]
Filling pellet	Bentonite & air	MX80 bentonite, 50 v.% air [37,42]
Hole buffer	Bentonite	MX80 bentonite [42]
Tunnel coating	Projected concrete	steel-reinforced [38,43–45]
Dense backfill	Bentonite & cement	50 v.% each [42,44]
Tunnel filling	Air	1.327 wt% water [46]
Outside of the tunnel	Continental crust rock	[47]
Flooding environment	Pure light water	[37]

OpenMC Cask Criticality Analysis of Novel TRISO-Pebble Cask Design

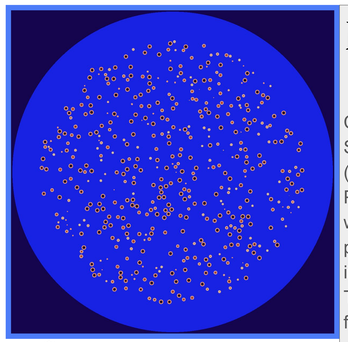
Matthew Disimone Zoey Hiiiiii

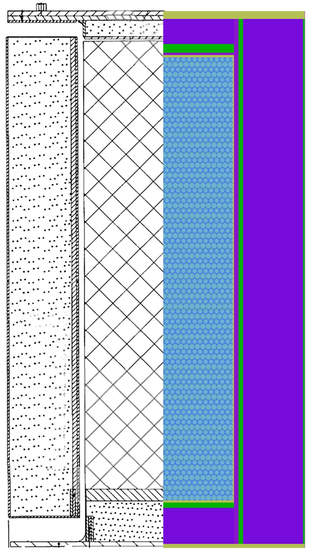
Introduction and Background

An alternative to the fuel rod assemblies are spherical Tri-Structural Isotropic (TRISO) fuel. TRISO fuel is attractive because of its higher burnup and use in pebble circulation systems, allowing certain HTGR designs to opperte for decades without shutdown for refueling. Additionally, the layers of TRISO pebbles prevent fuel failure and fission product leakage, adding an aspect of safety to design..

This work is motivated by the future production of pebble bed reactors without the proper construction of fuel containment casks for these types of pebbles. Work has been done (Paquette at el 2025) to study the behavior of adapting contemporary fuel cask designs. This work investigated stacking pebbles within the multi-purpose container (MPC) and filling the void space with high conductivity metals. However, this design raises issues in terms of retrieving the spent fuel for recycling and the physical constraints of molten metal. This work proposes a novel design that would make storage and retrieval of the these pebbles possible. This is achieved by using a corrugated, isometric structure of stainless steel doped with natural boron to pack the depleted pebbles in a uniform and repeating pattern. For mental clarity, this design was inspired by egg-cartons.

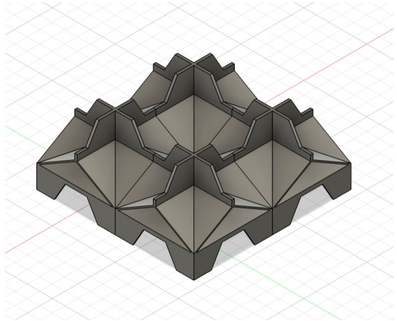
This work leverages the depleted pebble concentrations modeled from Richter, with OpenMC k-eigenvalue solvers to find the keff of the novel cask design's system to determine if such a design passes the NRC's hypothetical accident conditions in regards to criticality. The application of this work is a novel storage design that can be retro-fitted to fit into current MPC's

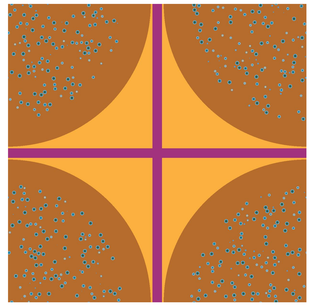


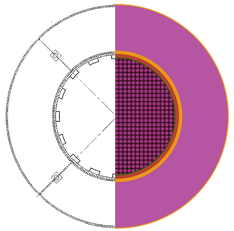


Methods

Cask geometry, including shield thickness, was based off a Holtec Hi-Storm 100 Dry Storage Cask. It features a 69.5-[in] radius and 190.5-[in] tall Multi-Purpose Container (MPC) where the storage. All shield and geometric components are tabulated in the Parts/Material/Reference table. The structure that holds the pebbles, the 'Fuel Basket,' was modeled in Fusion 360 and imported into OpenMC. A unit-cell of this repeating pattern is 6.25-[cm] across and 11-[cm] tall. There are 19,300 Triso kernels inside of an individual pebble. Pebble and the kernel geometry was based off X-energy's Triso-X fuel. The initial and depleted fuel compositions were provided by Richter, who modeled the fuel composition over 1500 days, the average lifetime of a pebble within an Xe-100. The original fuel composition before depletion was Uranium Oxycarbide enriched to 15%. Simulations either modeled the entire cask with shielding and geometry, or an infinite lattice of the BCC frame and pebbles inside an infinite lattice using periodic boundary conditions.

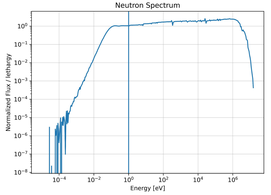






	OpenMC
Version	0.15.3 (Nov. 22 2025)
Cross Sections Library	ENDF/B-VIII.0
Particles per batch	20,000
Batches	100
Inactive Batches	20
Initial Source Angular Distribution	Isotropic
Boundary Conditions	Vacuum

Part(s)	Material	Specifications / Reference
Overpack Shell	Carbon Steel	ASTM A516 Grade 70
Radial Shield I	Concrete	Portland Concrete
Radial Shield II	Carbon Steel	ASTM A516 Grade 70
Base and Top Plate	Carbon Steel	ASTM A516 Grade 70
MPC Concrete Base	Concrete	Portland Concrete
MPC Steel base	Carbon Steel	ASTM A516 Grade 70
MPC	Stainless Steel	316 Stainless Steel
Pebble Basket	**Doped Stainless Steel**	**316 Stainless Steel doped w/ 1.0% natural boron (neutron absorber)**
Air Annulus	Air / Water (accident Scenario)	Natural Air composition / H2O (accident Scenario)
MPC 'Coolant'	Helium	Helium
Outside Cask Region	Air / Water (accident Scenario)	Natural Air composition / H2O (accident Scenario)
TRISO Pebble	Graphite	Graphite
TRISO Particle / Fuel Kernel	Layered kernel with depleted fuel composition [Richter]	PyC, SiC, Buffer, Depleted Fuel [Originally U^{235} Enriched to 13.86% w/o]



Conclusion

The BCC unit cell with periodic boundary conditions without doping is well below the k_eff NRC regulation for normal conditions. The BCC unit cell with perioding boundary conditions in the fully submerged HAC was _____.

Methods

Cask geometry, including shield thickness, was based off a Holtec Hi-Storm 100 Dry Storage Cask. It features a 69.5-[in] radius and 190.5-[in] tall Multi-Purpose Container (MPC) where the storage. All shield and geometric components are tabulated in the Parts/Material/Reference table. The structure that holds the pebbles, the 'Fuel Basket,' was modeled in Fusion 360 and imported into OpenMC. A unit-cell of this repeating pattern is 6.25-[cm] across and 11-[cm] tall. There are 19,300 Triso kernels inside of an individual pebble. Pebble and the kernel geometry was based off X-energy's Triso-X fuel. The initial and depleted fuel compositions were provided by Richter, who modeled the fuel composition over 1500 days, the average lifetime of a pebble within an Xe-100. The original fuel composition before depletion was Uranium Oxycarbide enriched to 15%. Simulations either modeled the entire cask with shielding and geometry, or an infinite lattice of the BCC frame and pebbles inside an infinite lattice using periodic boundary conditions.

Part(s)	Material	Specifications / Reference
Overpack Shell	Carbon Steel	ASTM A516 Grade 70
Radial Shield I	Concrete	Portland Concrete
Radial Shield II	Carbon Steel	ASTM A516 Grade 70
Base and Top Plate	Carbon Steel	ASTM A516 Grade 70
MPC Concrete Base	Concrete	Portland Concrete
MPC Steel base	Carbon Steel	ASTM A516 Grade 70
MPC	Stainless Steel	316 Stainless Steel
Pebble Basket	**Doped Stainless Steel**	**316 Stainless Steel doped w/ 1.0% natural boron (neutron absorber)**
Air Annulus	Air / Water (accident Scenario)	Natural Air composition / H_2O (accident Scenario)
MPC 'Coolant'	Helium	Helium
Outside Cask Region	Air / Water (accident Scenario)	Natural Air composition / H_2O (accident Scenario)
TRISO Pebble	Graphite	Graphite
TRISO Particle / Fuel Kernel	Layered kernel with depleted fuel composition [Richter]	PyC, SiC, Buffer, Depleted Fuel [Originally U_{235} Enriched to 13.86% w/o]

Conditions / Scenario	Model	**K_eff**
Normal Conditions (NC) No Boron Doping	BCC w/ periodic boundaries	0.34369 +/- 0.00031
Normal Conditions (NC) Boron Doping	BCC w/ periodic boundaries	0.07031 +/- 0.00005
Full Water Submersion (HAC) No Boron Doping	BCC w/ periodic boundaries	0.31468 +/- 0.00030
Full Water Submersion (HAC) Boron Doping	BCC w/ periodic boundaries	0.05726 +/- 0.00010
Normal Conditions (NC) Boron Doping	Full Cask w/ vacuum boundaries	0.05278 +/- 2e-5

Component	Geometry
Cask	Height : 231.75 [in] Radius: 66.25 [in]
Radial Shields	Thickness (Concrete) : 26.75 [in[Thickness (Steel) : 1.25 [in]
MPC	Height : 190.50 [in] Radius : 34.75 [in] Thickness : 0.50 [in]
BCC Unit Cell	Height : 4.33 [in] | 11 [cm] Width : 2.46 [in] | 6.25 [cm] Length : 2.46 [in] | 6.25 [cm]
Triso Pebble	Radius : 1.18 [in[| 6.00 [cm]
Triso Particle	Radius : 425 μm

Conditions / Scenario	Model	K_eff
Normal Conditions (NC) No Boron Doping	BCC w/ periodic boundaries	**0.34369** +/- **0.00031**
Normal Conditions (NC) Boron Doping	BCC w/ periodic boundaries	**0.07031** +/- **0.00005**
Water Submersion (HAC) No Boron Doping	BCC w/ periodic boundaries	**0.31468** +/- **0.00030**
Water Submersion (HAC) Boron Doping	BCC w/ periodic boundaries	**0.05726** +/- **0.00010**
Normal Conditions (NC) Boron Doping	Full Cask w/ vacuum boundaries	**0.05278** +/- **0.00002**

OpenMC Cask Criticality Analysis of Novel TRISO-Pebble Cask Design

Matthew Disimone, Zoe Richter, Kathryn Huff

Introduction

- **Triso Particles**: HALEU UCO fuel particles coated in protective layers of various carbides, ~1 mm wide.
- **Triso Pebbles:** 6cm diameter graphite spheres with approximately 19,000 TRISO particles embedded within. Advertise higher burnup combined with recirculation systems in HTGRs, allowing for extended periods without refueling outages.
- **Motivation:** Planned proper storage for spent pebbles in time for future pebble-bed HTGR deployment (Xe-100).
- **Past Work**: Adapting contemporary fuel cask designs by stacking pebbles within the multi-purpose container (MPC) and filling the void space with high-conductivity metals (Paquette et al. 2025).
- **This Work:** Proposes a novel fuel blanket design to hold the spent TRISO pebbles. Designed for future retrieval and without the physical constraint caused by the metal.

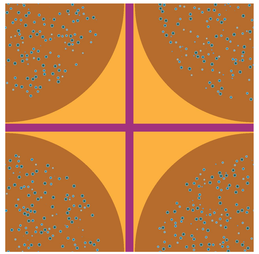


Figure 1: Cross section of the top stack in the BCC model. Magenta region is the fuel blanket .

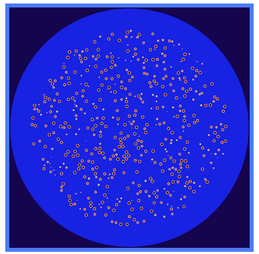


Figure 2: Cross section of individual TRISO-X pebble with particles visible.

	OpenMC
Version	0.15.3 (Nov. 22 2025)
Cross Sections Library	ENDF/B-VIII.0
Particles per batch	20,000
Batches	100
Inactive Batches	20
Initial Source Angular Distribution	Isotropic
Boundary Conditions	Periodic for BCC Vacuum for Full

Table 1: OpenMC build and simulation information.

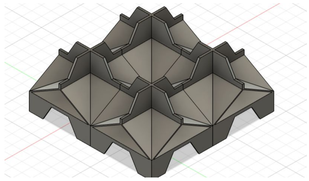


Figure 3: Fusion 360 CAD model of the fuel blanket / Pebble Holder.

Component	Geometry
Cask	Height : 231.75 [in] Radius: 66.25 [in]
Radial Shields	Thickness (Concrete) : 26.75 [in] Thickness (Steel) : 1.25 [in]
MPC	Height : 190.50 [in] Radius : 34.75 [in] Thickness : 0.50 [in]
BCC Unit Cell	Height : 4.33 [in] Width : 2.46 [in] Length : 2.46 [in]
Triso Pebble	Radius : 6.00 [cm]
Triso Particle	Radius : 425 [μm]

Table 2: Relevant geometry and parameters for the simulations.

Methods

- Cask Geometry based on a **Holtec Hi-Storm 100.**
- Pebble and the kernel geometry based on **X-energy's Triso-X.**
- Depleted fuel compositions were provided by Richter
  - OpenMC **Selective Iterative Depletion** over **1500 days**, the average lifetime of an Xe-100 pebble, over 4 iterations.
- Original fuel comp: **Uranium Oxycarbide** at 15% w/o
- **OpenMC** Simulations:
  - **Full cask** : Shielding with full geometry and leakage
  - **Body centered cubic (BCC) :** infinite lattice w/ periodic BC
  - **NC** : Normal conditions w/ inert helium coolant
  - **HAC** : Hypothetical accident condition w/ water submersion

Part(s)	Material	Specifications / Reference
Overpack Shell	Carbon Steel	ASTM A516 Grade 70
Radial Shield I	Concrete	Portland Concrete
Radial Shield II	Carbon Steel	ASTM A516 Grade 70
Base and Top Plate	Carbon Steel	ASTM A516 Grade 70
MPC Concrete Base	Concrete	Portland Concrete
MPC Steel base	Carbon Steel	ASTM A516 Grade 70
MPC	Stainless Steel	316 Stainless Steel
Pebble Basket	**Doped Stainless Steel**	**316 Stainless Steel doped w/ 1.0% natural boron (neutron absorber)**
Air Annulus	Air / Water (accident Scenario)	Natural Air composition / H2O (accident Scenario)
MPC 'Coolant'	Helium	Helium
Outside Cask Region	Air / Water (accident Scenario)	Natural Air composition / H2O (accident Scenario)
TRISO Pebble	Graphite	Graphite
TRISO Particle / Fuel Kernel	Layered kernel with depleted fuel composition [Richter]	PyC, SiC, Buffer, Depleted Fuel [Originally U^{235} Enriched to 15.0%]

Table 3: OpenMC regions & individual material compositions.

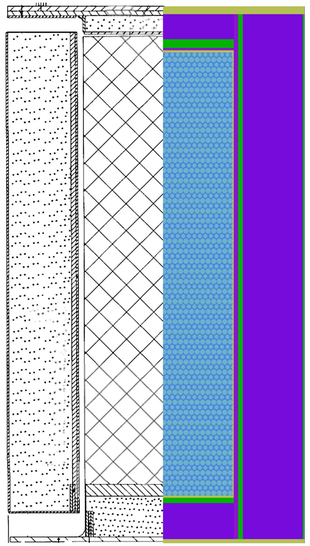


Figure 4: OpenMC cross sectional plots superimposed onto Holtec Hi-Storm 100 references.

Results

- NC BCC w/ helium: **Subcritical**
- NC BCC w/ Boron doped blanket : **Subcritical**
- HAC BCC w/ water submersion : **Subcritical**
  - Smaller keff than NC case
  - Neutron spectrum shows fissions, however water absorbance effect is much stronger
    - Overmoderation
- HAC BCC w/ Boron doped blanket : **Subcritical**
  - Lowest keff combination
- Full Cask w/ Boron doped blankets : **Subcritical**

To minimize criticality, a combination of water and boron-doped place should be used. **This first novel design meets the NRC's neutronics criteria for both normal and accident conditions. This work also demonstrates the passive safety mechanisms inherent to TRISO pebble designs.**

Conditions / Scenario	Model	K_eff
Normal Conditions (NC) No Boron	BCC Periodic	**0.34369** +/- **0.00031**
Normal Conditions (NC) w/ Boron	BCC Periodic	**0.07031** +/- **0.00005**
Water Submersion (HAC) No Boron	BCC Periodic	**0.31468** +/- **0.00030**
Water Submersion (HAC) w/ Doping	BCC Periodic	**0.05726** +/- **0.00010**
Normal Conditions (NC) w/ Boron	Full Cask Vacuum Leakage	**0.05278** +/- **0.00002**

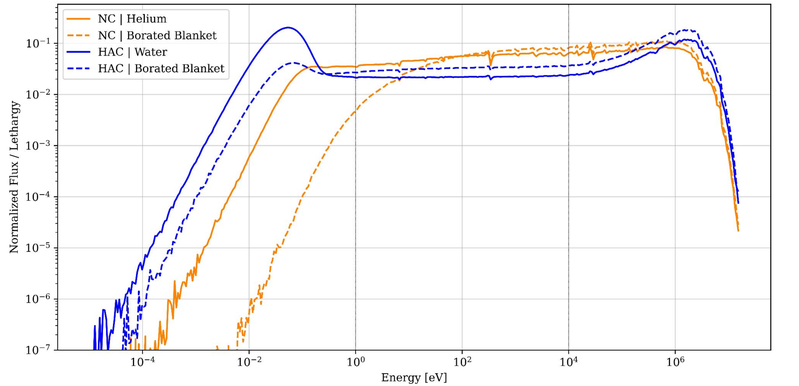
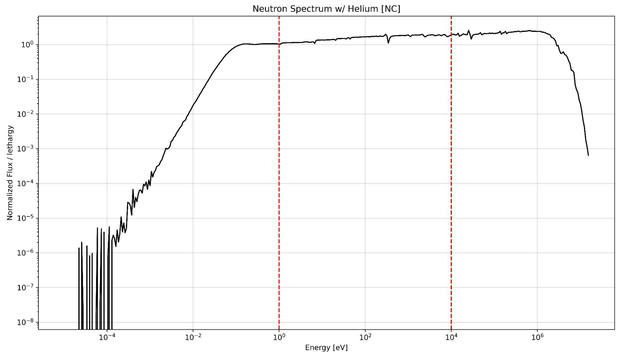
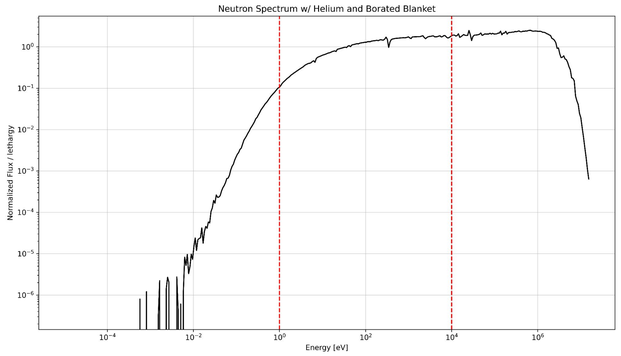
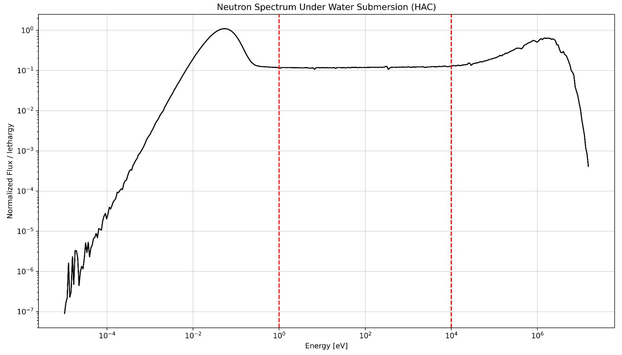
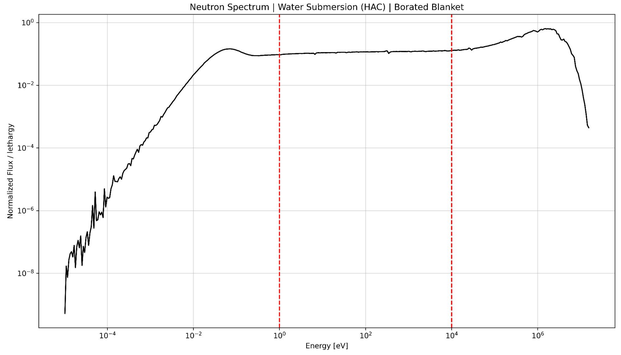


Figure 5: OpenMC Neutron Flux Spectra for BCC Simulations. Thermal and fast peaks for HAC suggest fission (Blue Lines). Boron poison effects shown in thermal and resonance regions (Dotted Lines).









Methods

- Cask Geometry based on a **Holtec Hi-Storm 100**.
- Pebble and the kernel geometry based on X-energy's **Triso-X.**
- Depleted fuel compositions were provided by Richter,
- Modeled the fuel composition over 1500 days, the average lifetime of a pebble within an Xe-100.
- The original fuel composition before depletion was Uranium Oxycarbide enriched to 15%.
- **OpenMC** Simulations:
  - Full cask with shielding and geometry
  - Infinite lattice of body centered cubic (BCC) infinite lattice with periodic boundary conditions.

Results

- NC BCC w/ helium: **Subcritical**
- NC BCC w/ Boron doped blanket : **Subcritical**
- HAC BCC w/ water submersion : **Subcritical**
  - Smaller keff than NC case
  - Neutron spectrum shows fissions, however water absorbance effect is much stronger
    - Overmoderation
- HAC BCC w/ Boron doped blanket : **Subcritical**
  - Lowest keff combination

To minimize criticality, a combination of water and boron-doped place should be used. **This first novel design meets the NRC's neutronics criteria for both normal and accident conditions,** demonstrating the redundant and passive safety mechanisms inherent to the TRISO pebble design.

Methods

Cask geometry was based on a Holtec Hi-Storm 100 Dry Storage Cask. Pebble and the kernel geometry were based on X-energy's Triso-X fuel. Depleted fuel compositions were provided by Richter, who modeled the fuel composition over 1500 days, the average lifetime of a pebble within an Xe-100. The original fuel composition before depletion was Uranium Oxycarbide enriched to 15%. Simulations either modeled the entire cask with shielding and geometry, or an infinite lattice of the BCC frame and pebbles inside an infinite lattice using periodic boundary conditions.